

Aligning Large Language Models with Reinforcement Learning from Human Feedback

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ABSTRACT

Large language models can produce untruthful, toxic, or misaligned outputs when trained solely on unsupervised next-token prediction objectives. We apply Reinforcement Learning from Human Feedback (RLHF) using proximal policy optimization (PPO) and reward modeling. Fine-tuning foundation models with human preference rankings significantly improves safety, truthfulness, and conversational helpfulness.

Keywords: Reinforcement Learning, RLHF, Large Language Models, Alignment, Prompt Engineering

1. Introduction

Pretrained language models excel at completing prompts, but their objective does not match human intent. Aligning model behaviors with human ethical values and instruction following requires explicit preference optimization.

2. RLHF Methodology

We collect comparison data of model outputs labeled by human annotators. A reward model is trained to predict preferred responses, and the policy model is fine-tuned against this reward using PPO with a Kullback-Leibler penalty to prevent policy drift.

3. Empirical Results

Human evaluators strongly preferred RLHF-aligned models over supervised baseline models by 85%. Toxicity metrics dropped by 60% while preserving general conversational capabilities.

4. Conclusion

Reinforcement learning from human feedback provides an effective framework for aligning generative artificial intelligence systems with human values and safety standards.